AI-Powered Kitchen Ecosystem for Food Waste Reduction and Nutrition Optimization

Perera U M K
Department of information Technology
Sri Lanka Institute of information Technology
Malabe, Sri Lanka
it22131942@my.sliit.lk

Shahmi M T M
Department of information Technology
Sri Lanka Institute of information Technology
Malabe, Sri Lanka
it22083982@my.sliit.lk

Muraleswaran D
Department of information Technology
Sri Lanka Institute of information Technology
Malabe, Sri Lanka
it22339010@my.sliit.lk

Jayasundara D H
Department of information Technology
Sri Lanka Institute of information Technology
Malabe, Sri Lanka
it22117946@my.sliit.lk

Ms. Lokesha Weerasinghe
Department of information Technology
Sri Lanka Institute of information Technology
Malabe, Sri Lanka
lokesha.w@sliit.lk

Ms. Chathurya Prabhavi Kumarapperuma
Department of information Technology
Sri Lanka Institute of information Technology
Malabe, Sri Lanka
chathurya.k@sliit.lk

***Abstract*— Food waste and nutritional imbalance remain major global challenges affecting environmental sustainability, economic efficiency, and public health. Although several digital applications exist for recipe recommendation, nutrition monitoring, and grocery shopping, most operate independently and fail to support the entire household food lifecycle. This study proposes an Artificial Intelligence Powered Kitchen Ecosystem that integrates cooking assistance, nutritional guidance, personalized food expiry prediction, and intelligent grocery recommendation within a unified framework. The system employs semantic recipe retrieval, machine learning–based nutritional deficiency prediction, regression-based shelf-life estimation, and hybrid recommendation algorithms for grocery suggestions. The integrated architecture enables continuous data exchange between modules, allowing contextual decision support across food storage, preparation, and consumption stages. Experimental results demonstrate strong predictive performance. The cooking assistant achieved 86.84% classification accuracy, the nutritional guidance module achieved 88.12% accuracy in predicting micronutrient deficiency risk, and the food expiry prediction system achieved an R squared value of 0.7443. These results indicate that integrated artificial intelligence systems can significantly improve household food management and promote sustainable consumption.**

***Keywords*— Artificial Intelligence, Food Waste Reduction, Machine Learning, Nutritional Risk Prediction, Shelf-Life Prediction, Recommender Systems**

I. Introduction

Food waste and nutritional imbalance represent two major challenges affecting modern food systems. According to global food system reports, nearly one-third of food produced worldwide is lost or wasted throughout the supply chain, with a significant proportion occurring at the household level [1]. Inefficient food management practices such as poor meal planning, inaccurate shelf-life estimation, and unnecessary grocery purchasing contribute to this problem. At the same time, nutritional deficiencies continue to affect millions of people due to lack of dietary awareness and limited access to personalized nutritional guidance [2].

The rapid growth of digital technologies has led to the development of several applications designed to assist users in food-related decision making. Recipe recommendation platforms help users discover meals based on ingredients, nutrition tracking applications monitor dietary intake, and grocery shopping systems recommend products based on purchasing behavior. However, these solutions typically function as isolated systems without interaction between cooking, storage, nutrition monitoring, and purchasing processes. As a result, users must manually coordinate between different platforms when planning meals, managing food inventory, and purchasing groceries.

Furthermore, many existing recipe recommendation systems rely on simple keyword matching or rule-based approaches that cannot effectively capture the semantic relationships between ingredients and recipes. Advances in natural language processing have introduced semantic embedding models that enable contextual understanding of ingredient descriptions and cooking instructions [3]. These models allow systems to retrieve recipes based on conceptual similarity rather than exact keyword matching.

Artificial intelligence techniques have also been widely applied in nutrition monitoring and health risk prediction. Machine learning algorithms such as Random Forest and

gradient boosting models have demonstrated strong performance in identifying dietary patterns and predicting nutritional deficiencies [4]. Similarly, predictive models have been used to estimate food shelf life under varying environmental conditions using regression techniques capable of capturing nonlinear relationships between storage parameters and spoilage rates [5].

In addition, recommendation systems have evolved significantly in recent years through the development of hybrid recommendation architectures that combine collaborative filtering and content-based approaches to improve recommendation accuracy [6]. Despite these advances, most food-related recommendation systems do not incorporate contextual information such as household inventory levels, nutritional requirements, or food expiry status.

This research addresses these limitations by proposing an AI-Powered Kitchen Ecosystem that integrates cooking assistance, nutritional risk prediction, personalized food expiry prediction, and intelligent grocery recommendation within a unified architecture. By enabling continuous interaction between these modules, the system provides adaptive decision support across the entire household food lifecycle.

II. Literature Review

The challenge of intelligent food management has generated substantial research across food computing, recommender systems, and digital health domains. However, these advancements have largely progressed independently, resulting in specialized yet fragmented solutions. The following review synthesizes key developments across recipe intelligence, nutritional modeling, expiry prediction, and grocery recommendation, highlighting their limitations and motivating the need for integrated ecosystem-level architectures.

Recent research has explored intelligent food management systems aimed at improving cooking assistance, dietary monitoring, and food waste reduction. However, most studies focus on individual functionalities rather than developing a comprehensive ecosystem capable of addressing the entire food lifecycle.

Morol et al. proposed a deep learning–based recipe recommendation system that detects ingredients from food images and suggests recipes using convolutional neural networks [7]. Their study demonstrated that visual ingredient detection can improve the effectiveness of cooking assistance applications. However, the system primarily focuses on ingredient recognition and does not incorporate nutritional evaluation or food storage management.

With the advancement of multimodal artificial intelligence, recent research has explored the use of large language models for culinary tasks. Mohbat and Zaki introduced LLaVA-Chef, a multimodal generative model capable of generating cooking instructions from food images and textual descriptions [8]. While this approach demonstrates the potential of multimodal models for cooking applications, generated recipes may lack contextual awareness of household ingredients and dietary requirements.

Semantic retrieval models have also been applied to improve recipe recommendation accuracy. Sentence-BERT architectures enable contextual sentence embeddings capable of capturing semantic similarity between ingredient descriptions and recipe instructions [3]. These models have shown improved performance compared with traditional keyword-based retrieval methods. Nevertheless, most implementations focus only on recipe retrieval and do not integrate broader food management functionalities such as inventory tracking or expiry prediction.

In the field of nutrition monitoring, machine learning algorithms have been widely applied to predict dietary risk and health outcomes. Random Forest classifiers have demonstrated strong performance in identifying nutritional deficiency risks and modeling dietary patterns due to their ability to handle complex nonlinear relationships within nutritional datasets [4]. Recent studies have highlighted the effectiveness of machine learning techniques in analyzing nutritional intake and predicting potential health risks.

Food shelf-life prediction has also received increasing attention in recent research. Machine learning approaches such as gradient boosting and deep learning models have been used to predict food spoilage by analyzing environmental conditions and storage parameters [5]. These predictive models provide valuable insights for reducing food waste by enabling more accurate shelf-life estimation.

Recommender systems for grocery shopping have evolved from simple collaborative filtering models to hybrid recommendation architectures capable of incorporating contextual information such as user preferences, dietary restrictions, and purchasing behavior. Hybrid recommendation approaches combining collaborative filtering and content-based filtering have demonstrated improved recommendation accuracy and robustness in e-commerce environments [6].

Despite these advancements, most existing studies focus on specific applications such as recipe recommendation, nutrition analysis, or shopping recommendation. Few systems integrate these functionalities within a single intelligent framework capable of supporting the entire household food lifecycle. The proposed AI-Powered Kitchen Ecosystem addresses this research gap by combining cooking assistance, nutritional guidance, food expiry prediction, and grocery recommendation within a unified artificial intelligence architecture.

III. System Overview Architecture

The AI-Powered Kitchen Ecosystem follows a modular architecture designed to integrate multiple artificial intelligence components within a unified framework. The overall system architecture is illustrated in Fig. 2.

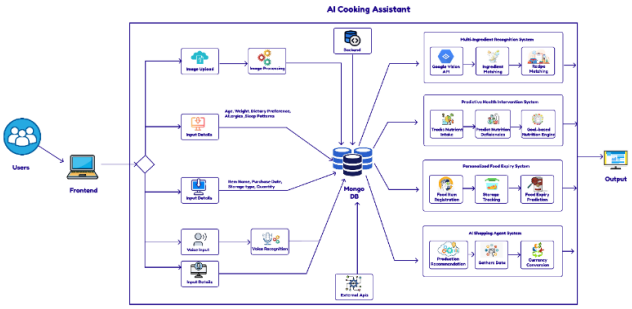

Fig. 2. Integrated system architecture of the AI-Powered Kitchen Ecosystem

The system consists of three primary layers: a presentation layer, an application orchestration layer, and a shared data layer. The presentation layer provides the user interface through which users interact with the system. Users can enter ingredients through text queries, upload images of available food items, track nutritional intake, and request grocery recommendations. The interface communicates with backend services through REST-based APIs.

The application orchestration layer integrates four core intelligent modules. The spontaneous cooking assistant analyzes available ingredients and retrieves relevant recipes using semantic similarity models. The nutritional guidance module evaluates dietary intake data and predicts micronutrient deficiency risks using machine learning classification algorithms. The personalized food expiry prediction module estimates shelf life of stored food items using regression models that incorporate environmental and storage conditions. The smart shopping agent generates grocery recommendations using hybrid ranking techniques that consider dietary requirements and inventory status.

The shared data layer functions as a centralized repository for storing food inventory records, nutritional intake data, user preferences, and historical purchase information. Continuous data exchange between modules allows contextual interactions where cooking recommendations influence nutritional monitoring and shopping suggestions, enabling a closed-loop food management ecosystem.

IV. Methodology

A. *Spontaneous Cooking Assistant with Vision-Based Ingredient Detection*

The Spontaneous Cooking Assistant was designed to support ingredient-based recipe discovery through a hybrid pipeline combining semantic recipe classification, fuzzy ingredient matching, vision-based ingredient detection, and multilingual translation support. The module was developed using a curated trilingual Sri Lankan recipe corpus, an ingredient dictionary with synonym mappings, a food image dataset, and a user validation dataset collected during a three-week intervention study. These datasets enabled semantic retrieval, culturally adaptive ingredient recognition, and empirical validation of food waste reduction.

Prior to modeling, duplicate recipes were removed and textual data were normalized through lowercase conversion, whitespace trimming, and standardized formatting. Recipe representations were generated using Sentence-BERT (all-MiniLM-L6-v2), which produced 384-dimensional dense embeddings capturing semantic relationships among ingredients and preparation context. In addition to semantic embeddings, TF-IDF features and structured attributes such as recipe category, difficulty level, ingredient count, and cooking time were used to strengthen classification performance. For fuzzy ingredient matching, a Levenshtein similarity matrix was constructed across the ingredient vocabulary, and threshold optimization identified 0.8 as the most effective similarity cutoff. For image-based ingredient detection, food images were standardized to 224 × 224 resolution and processed using Google Vision API with data augmentation to improve robustness across ingredient categories.

The semantic classification model was implemented using Sentence-BERT as the embedding backbone followed by a supervised classification head consisting of a 384-to-128 fully connected layer, ReLU activation, dropout regularization, and a final softmax output layer for seven recipe categories. The model was trained using Adam optimization with categorical cross-entropy loss, a batch size of 16, and early stopping to reduce overfitting. Fuzzy matching and image detection were integrated with the semantic classification pipeline to improve performance under lexical variation and visual ambiguity.

Evaluation was conducted using stratified five-fold cross-validation to preserve minority recipe category representation. The module achieved 86.84% classification accuracy with macro-F1 of 0.862, while fuzzy ingredient matching achieved 75% accuracy at the optimized threshold and the vision-based detection component achieved 87.8% accuracy. A three-week user study involving 15 participants further showed a statistically significant 73.9% reduction in weekly household food waste, demonstrating the real-world effectiveness of the cooking assistant.

B. *Nutritional Guidance – Predictive Health Intervention System*

The Nutritional Guidance component was developed to predict micronutrient deficiency risk from dietary intake patterns using requirement-normalized nutritional modeling. The module integrates food composition data, age-specific nutrient requirement standards, health-condition adjustment factors, and temporal intake logs to support preventive health intervention. After preprocessing and synthetic intake generation, 3,200 labeled dietary samples were constructed

with balanced representation across LOW, MEDIUM, and HIGH deficiency risk categories.

Preprocessing involved data cleaning, type conversion, aggregation of dietary intake records, and computation of requirement-normalized adequacy ratios for energy, protein, calcium, and iron. These ratios transformed raw intake values into interpretable deficiency signals relative to recommended thresholds. Risk labels were generated using a rule-based adequacy framework in which cumulative nutrient deficiency indicators were mapped to LOW, MEDIUM, and HIGH risk classes. This formulation enabled supervised learning in the absence of laboratory biomarker labels.

The predictive model was implemented using a Random Forest classifier due to its robustness in handling structured nutritional data and nonlinear threshold relationships between nutrient intake and deficiency risk. Each training instance included both absolute nutrient totals and normalized adequacy ratios, specifically age, total energy, total protein, total calcium, total iron, ratio of energy adequacy, ratio of protein adequacy, ratio of calcium adequacy, ratio of iron adequacy, and a binary health-condition indicator. The model was configured with 250 trees, maximum depth of 10, balanced class weighting, and a fixed random seed for reproducibility.

Training data were generated by simulating realistic daily dietary profiles using combinations of food items and portion sizes. The dataset was partitioned using an 80:20 stratified train-test split, and five-fold cross-validation was applied to assess generalization stability. The model achieved 88.12% test accuracy and a mean cross-validation accuracy of 89.31% with a standard deviation of ±1.44%. Class-wise precision and recall ranged from 0.86 to 0.90, with a macro-F1 score of approximately 0.88 and macro-average AUC of approximately 0.91, indicating strong discriminative performance across risk categories.

C. Personalized Food Expiry Prediction

The Personalized Food Expiry Prediction component was developed to estimate adaptive yet biologically safe shelf life under varying storage conditions. The problem was formulated as a supervised nonlinear regression task in which the target variable was the number of days until expiry. Two complementary datasets were used: a food expiry prediction dataset containing historical records of food items, storage conditions, environmental attributes, and observed expiry duration, and a base expiry reference dataset containing biologically safe shelf-life limits for pantry, refrigerator, and freezer storage. After controlled cleaning, 2,048 records were retained for modeling.

Preprocessing included removal of duplicate and inconsistent records, standardization of storage labels, conversion of expiry information into numerical shelf-life duration, and extraction of contextual features including item type, category, storage type, temperature, humidity, quantity, and base expiry. Exploratory analysis showed that storage type and food category were strong predictors, while temperature had a moderate negative relationship with expiry duration.

The predictive model was implemented using CatBoostRegressor because of its ability to handle categorical variables effectively and capture nonlinear spoilage dynamics. The final configuration used 600 boosting iterations, tree depth of 6, learning rate of 0.05, Mean Absolute Error as the loss function, and early stopping. The dataset was divided using an 80:20 train-test split, and five-fold cross-validation was applied to evaluate generalization stability.

To ensure food safety, a validation layer was introduced after regression. The raw prediction was constrained using a lower-bound safety rule in which predicted shelf life could not fall below 60% of the corresponding biologically defined base expiry. A global bound of 0.5× to 1.5× base expiry was further applied to prevent unrealistic underestimation or unsafe extension. To support personalization, an Adaptive Expiry Adjustment mechanism updated future predictions using user feedback, while a Smart Consumption Priority score translated validated expiry estimates into actionable urgency levels for household consumption planning.

The model achieved an R^2 of 0.7443 and MAE of 0.0960 days, with five-fold cross-validation producing mean R^2 of 0.742 ± 0.012 and mean MAE of 0.101 ± 0.008 days. These results indicate that the selected framework provides accurate, stable, and safety-aware expiry prediction suitable for real-world household food management.

D. AI-Powered Smart Shopping Agent with Trilingual Support

The Smart Shopping Agent was developed to support real-time price comparison, personalized recommendation, and multilingual grocery search within the household food management ecosystem. The module integrated two main data sources: a real-time product dataset retrieved through the SerpAPI Google Shopping Engine and a user behavioral dataset derived from MongoDB interaction logs. The real-time product stream provided dynamic product attributes such as title, price, merchant, rating, and delivery metadata, while the behavioral dataset contained more than 1,200 user interactions across English, Sinhala, and Tamil queries, enabling preference modeling and trilingual recommendation.

Preprocessing included normalization of price and weight formats, caching of API responses, and translation of Sinhala and Tamil queries into English using a Sri Lanka–specific dictionary layer. Context-aware intent extraction was performed using Gemini-1.5-Flash, and TF-IDF vectorization was applied to user queries and product titles to support local content-based preference modeling. Additional contextual attributes such as shelf life, storage

tips, and wastage risk were generated to improve sustainability-aware ranking.

The recommendation engine employed a hybrid architecture consisting of a local behavioral model and a real-time large language model re-ranking layer. The local model used TF-IDF with cosine similarity to construct a user preference profile from historical search behavior and rank candidate products. The top-ranked items were then re-ranked using Gemini-1.5-Flash based on contextual factors including wastage risk and sustainability relevance. Final recommendation scores were computed using a weighted fusion strategy that assigned 60% weight to local behavioral similarity and 40% weight to contextual re-ranking.

Evaluation was conducted using a time-based 80:20 split with five-fold cross-validation to preserve temporal realism and reduce information leakage. The hybrid model was compared against Random, Popularity, and pure Content-Based baselines. It achieved HR@10 = 0.78, Precision@10 = 0.65, and NDCG@10 = 0.74, with statistically significant improvement over the Content-Based baseline ($p < 0.001$, Cohen's d = 0.82). Independent testing of the local TF-IDF model further showed approximately 85% category-level preference identification accuracy and inference latency below 50 milliseconds, confirming its usefulness as a low-latency fallback under constrained connectivity.

V. Results And Analysis

The experimental evaluation of the AI-Powered Kitchen Ecosystem was conducted independently for each subsystem using controlled validation protocols and statistically grounded performance metrics. The results demonstrate that the integrated architecture achieves consistent predictive accuracy, stable generalization, and measurable real-world impact across the household food lifecycle.

A. *Spontaneous Cooking Assistant*

The semantic recipe classification model achieved a mean accuracy of 86.84% under stratified 5- fold cross-validation, with a standard deviation of 1.71%, indicating stable generalization across folds. The macro-averaged F1-score was 0.862 and the weighted F1-score was 0.868, confirming balanced performance across minority and majority recipe categories. The 95% confidence interval for overall accuracy was [82.1%, 91.5%], demonstrating statistically reliable classification performance. Baseline comparisons revealed that the Sentence-BERT–based classifier outperformed keyword matching (62%), Logistic Regression (70.3%), and TF-IDF + SVM (74.8%) by a substantial margin, confirming the effectiveness of contextual semantic embeddings for low-resource culinary datasets.

The fuzzy ingredient matching component achieved 75% overall accuracy with an AUC of 0.91 at the optimized similarity threshold of 0.8. A significant inverse correlation ($r = -0.72$, $p < 0.001$) was observed between ingredient count and matching accuracy, indicating increased complexity in multi-ingredient queries. Vision-based ingredient detection reached 87.8% overall accuracy across six categories, with vegetables and spices exceeding 90%. These findings validate the hybrid design combining semantic embeddings, fuzzy lexical matching, and vision inference.

The real-world user interventi on study further demonstrated practical impact. Weekly household food waste decreased from 2.3 kg (SD = 0.5) at baseline to 0.6 kg (SD = 0.2) after three weeks, representing a 73.9% reduction. A paired t-test yielded $t(14) = 12.34$, $p < 0.001$, with Cohen's d = 3.18, indicating an extremely large effect size. These results confirm that the assistant not only performs accurately in controlled evaluation but also produces statistically significant real-world waste reduction.

Fig. 2. shows Overall matching accuracy reached 75% (95% CI [71%,79%]). An inverse correlation was observed between ingredient count and matching accuracy ($r = -0.72$, $p < 0.001$). Queries with 3 ingredients achieved 89%, declining to 67% for 10+ ingredients. Average response time was 427 ms (SD=89)

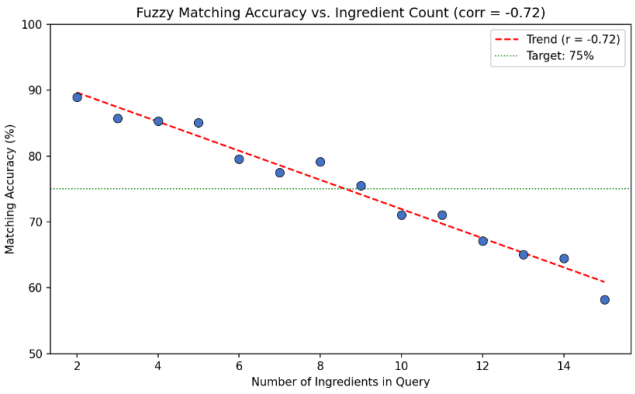

Fig.2 . Fuzzy Matching Accuracy vs Ingredient Count

B. *Nutritional Guidance – Predictive Health Intervention*

The Random Forest classifier achieved 88.12% accuracy on the held-out test set and a mean cross-validation accuracy of 89.31% (±1.44%) under stratified 5-fold validation. Precision and recall ranged between 0.86 and 0.90 across LOW, MEDIUM, and HIGH deficiency risk categories, with a macro-F1 score of approximately 0.88. Multi-class ROC analysis yielded a macro-average AUC of 0.91, demonstrating strong separability between risk levels.

Baseline comparisons against Logistic Regression, Decision Tree, and SVM classifiers showed inferior stability and reduced nonlinear modeling capacity, confirming the advantage of ensemble tree aggregation for adequacy-threshold modeling. The ablation study further demonstrated that incorporating requirement-normalized adequacy ratios significantly improved predictive consistency compared to raw intake totals alone, reinforcing the importance of normalization in preventive nutritional modeling.

Feature importance analysis revealed that total_calcium_mg, total_protein_g, and total_iron_mg were dominant predictors, while adequacy ratios enhanced classification robustness when combined with absolute intake values. These findings validate the hybrid feature representation strategy integrating magnitude-sensitive and requirement-normalized signals.

Fig. 3. the confusion matrix shows strong diagonal dominance. Minor misclassifications occur primarily between MEDIUM and HIGH risk, reflecting borderline

adequacy thresholds rather than structural bias. Class-wise precision and recall ranged between 0.86 and 0.90, confirming balanced discrimination

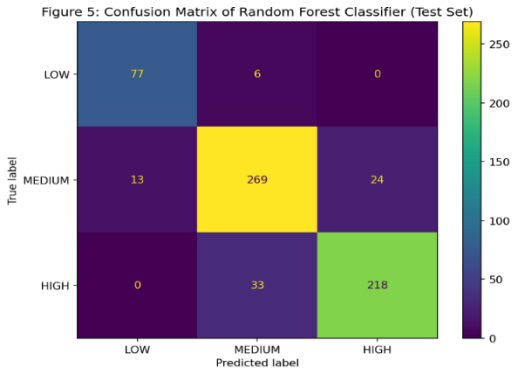

Fig. 3. Confusion Matrix

C. Personalized Food Expiry Prediction

The CatBoostRegressor achieved an R^2 score of 0.7443 and a Mean Absolute Error (MAE) of 0.0960 days (approximately 2.3 hours average deviation), indicating high predictive precision in shelf-life estimation. Five-fold cross-validation produced a mean R^2 of 0.742 (±0.012) and a mean MAE of 0.101 (±0.008), confirming stable generalization across partitions. Comparative baseline evaluation showed that Linear Regression ($R^2 \approx 0.41$), Random Forest ($R^2 \approx 0.48$), and MLP ($R^2 \approx 0.46$) were substantially outperformed by gradient boosting, justifying model selection.

Exploratory analysis demonstrated that storage type and food category were dominant predictive factors, while temperature exhibited moderate negative correlation (≈ −0.28) with expiry duration. Residual distribution analysis revealed symmetric error patterns without structural bias across storage conditions, although perishable categories exhibited slightly higher variance.

The integration of the safety validation layer ensured that predictions remained within biologically plausible bounds (0.5× to 1.5× base expiry), with the 0.60 lower-bound rule preventing unsafe underestimation. Empirical ratio analysis confirmed that most predictions clustered near the base expiry reference, indicating controlled personalization rather than unsafe extension. The Smart Consumption Priority mechanism further classified 19% of items as high urgency, 27% as medium urgency, and 54% as low urgency, reflecting realistic household storage dynamics.

The Fig. 4. presents Python-generated boxplots comparing pantry, refrigerator, and freezer storage. Freezer storage exhibits the highest median expiry duration, followed by refrigerator and pantry storage. Pantry storage demonstrates higher variability due to heterogeneous mixtures of perishable and shelf-stable foods. This visualization justified explicit inclusion of storage type as a high-importance categorical feature.

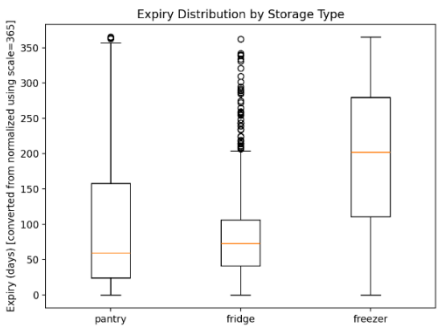

Fig. 4. Expiry Distribution by Storage Type

D. AI-Powered Smart Shopping Agent

The hybrid recommendation model achieved HR@10 = 0.78, Precision@10 = 0.65, and NDCG@10 = 0.74 under time-based evaluation. Compared to Random, Popularity, and pure Content-Based baselines, the hybrid architecture consistently demonstrated superior ranking quality. A paired t-test comparing the Content-Based baseline with the Hybrid model over N = 250 test interactions yielded $p < 0.001$ with Cohen's d = 0.82, indicating a large and statistically significant improvement. The 95% confidence interval for HR@10 performance was [0.768, 0.792], demonstrating low variance and stable ranking behavior.

Independent evaluation of the local TF-IDF component confirmed approximately 85% category-level preference identification accuracy with inference latency below 50 milliseconds, while hybrid API-dependent inference averaged approximately 1.5 seconds. This confirms that the architecture balances computational efficiency with contextual reasoning capability.

The Fig. 5. demonstrates significant improvement in Sinhala and Tamil query success rates after implementing the custom translation layer. Success improved from approximately 20% to above 90%, matching English query performance

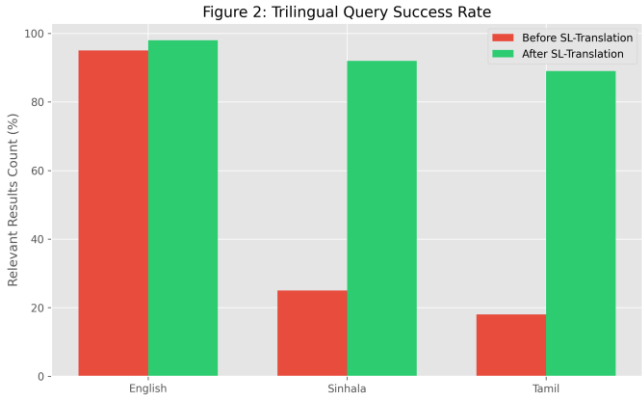

Fig. 5. Trilingual Query Success Rate

VI. Discussion

The experimental results demonstrate that the proposed AI-Powered Kitchen Ecosystem achieves reliable predictive performance across multiple artificial intelligence tasks while maintaining a unified and safety-aware architecture. Unlike isolated food recommendation or nutrition tracking applications, the proposed system integrates cooking assistance, nutritional risk prediction, food expiry estimation, and intelligent grocery recommendation within a closed-loop ecosystem. This integration enables cross-module reinforcement, where outputs from one subsystem provide contextual signals for others, improving decision support across purchasing, storage, preparation, and health monitoring stages.

The Spontaneous Cooking Assistant results confirm that semantic embedding techniques significantly outperform

traditional keyword-based approaches for low-resource culinary datasets. The large and statistically significant reduction in household food waste observed during the intervention study demonstrates that intelligent ingredient utilization can produce measurable behavioral impact beyond recommendation accuracy alone.

The Nutritional Guidance module shows that requirement-normalized adequacy modeling improves classification robustness compared with raw intake-based approaches. The high ROC-AUC and stable cross-validation results indicate that ensemble learning methods can reliably predict nutritional deficiency risk while supporting preventive health awareness.

The Personalized Food Expiry Prediction component highlights the importance of combining machine learning with biologically grounded safety constraints. The integration of lower-bound validation rules, bounded prediction ranges, and adaptive personalization mechanisms ensures that predictive outputs remain safe and practically applicable in real household environments.

Similarly, the Smart Shopping Agent demonstrates that hybrid recommendation architectures combining behavioral modeling with contextual large language model reasoning provide stronger ranking performance than single-paradigm systems.

Despite these promising results, several limitations remain. Some modules rely on partially synthetic or limited-scale datasets, particularly for nutritional intake modeling. The recipe dataset used by the cooking assistant is relatively small, which may limit scalability to broader culinary domains. Environmental variables in the expiry prediction module are currently static rather than sensor-driven, and the shopping agent does not yet incorporate collaborative filtering for cross-user preference learning. In addition, real-world validation was conducted over limited evaluation periods, and longer longitudinal studies across larger and more diverse user populations are required to confirm sustained impact.

Overall, while the proposed ecosystem demonstrates strong predictive capability and practical relevance, future work should focus on expanding datasets, integrating sensor-based environmental monitoring, incorporating collaborative recommendation strategies, and conducting large-scale real-world deployments.

VII. Conclusion and Futurework

This research presented an integrated AI-Powered Kitchen Ecosystem designed to address household food waste, nutritional deficiency risk, and purchasing inefficiency through a unified modular architecture. The system integrates four interoperable AI components: semantic recipe intelligence, predictive nutritional risk classification, safety-aware expiry regression with adaptive personalization, and hybrid multilingual shopping recommendation.

Comprehensive experimental evaluation demonstrated statistically significant performance improvements across classification, regression, and ranking tasks. The Spontaneous Cooking Assistant achieved stable semantic classification accuracy and demonstrated real-world food waste reduction. The Nutritional Guidance module produced high deficiency risk discrimination with strong ROC-AUC performance. The Personalized Food Expiry Prediction framework achieved low-error regression accuracy while enforcing biological safety constraints. The Smart Shopping Agent demonstrated statistically significant ranking improvement using hybrid contextual modeling.

The primary contribution of this work lies in demonstrating that predictive modeling, when combined with safety validation, personalization mechanisms, multilingual processing, and statistically rigorous evaluation, can be integrated into a cohesive ecosystem addressing the full household food lifecycle. Rather than isolated AI solutions, the proposed architecture establishes a closed-loop intelligent framework that connects purchasing, storage, preparation, and nutritional monitoring.

Future work will focus on expanding real-world deployment, incorporating IoT-based environmental sensing, integrating collaborative filtering mechanisms, and conducting larger-scale longitudinal validation studies. By advancing toward adaptive, safety-aware, and contextually intelligent household systems, this research contributes toward sustainable consumption practices and data-driven preventive health intervention.

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